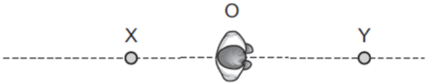
17	**C** Since loudspeakers emit waves that are in antiphase, the condition for waves to meet in phase is path difference = $(n + ½)\lambda$	**D**

For Options A and C, since OY < XY, observer O is in between X and Y as shown.



For Option A, OX = XY – OY = 3.70 – 1.25 = 2.45 m
Path difference of the waves from their respective sources to the observer
= OX – OY = 2.45 – 1.25 = 1.20 m = 2 λ, so not correct.

For Option C, OX = XY – OY = 2.50 – 2.00 = 0.50 m
Path difference of the waves from their respective sources to the observer
= OY – OX = 2.00 – 0.50 = 1.50 m = 2.5 λ, so correct.

For Options B and D, since OY > XY, observer is to the left of X as shown in original diagram.

Path difference of the waves from their respective sources to the observer = XY
Option B: Path difference = 1.20 = 2 λ, so incorrect.
Option D: Path difference = 1.60 m = 2.7 λ, so incorrect.

18	D	D